01. Introduction

Colorectal cancer, also known as colon or rectal cancer, develops in the large intestine (colon) or rectum. It usually begins as small growths called polyps, which may become cancerous over time if left untreated.

Early detection through screening significantly improves survival and treatment outcomes.

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02. What is Colorectal Cancer?

Colorectal cancer occurs when abnormal cells grow uncontrollably in the lining of the colon or rectum.

It often develops slowly over several years, beginning as non-cancerous polyps.

Cancer can spread through:

- **Nearby tissues**
- **Lymph nodes**
- **Bloodstream**
- **Liver**
- **Lungs**

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03. Types of Colorectal Cancer

The major types include:

Adenocarcinoma

- **Most common type**
- **Accounts for approximately 95% of cases**
- **Begins in mucus-producing glands**

Other Rare Types

- **Carcinoid tumors**
- **Gastrointestinal stromal tumors (GIST)**
- **Lymphomas**
- **Sarcomas**

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04. Risk Factors

Several factors increase the risk of colorectal cancer.

Non-Modifiable Factors

- **Age over 50 years**
- **Family history**
- **Genetic syndromes**
- **Personal history of colon polyps**
- **Inflammatory bowel disease**

Lifestyle Risk Factors

- **Low-fiber diet**
- **High consumption of red meat**
- **Processed foods**
- **Obesity**
- **Physical inactivity**
- **Smoking**
- **Excessive alcohol intake**

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05. Common Symptoms

Early colorectal cancer may not cause symptoms.

As cancer grows, symptoms may include:

- **Blood in stool**
- **Persistent diarrhea**
- **Constipation**
- **Narrow stools**
- **Abdominal pain**
- **Bloating**
- **Fatigue**
- **Unexplained weight loss**
- **Feeling that the bowel does not empty completely**
- **Iron deficiency anemia**

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06. Screening

Regular screening helps detect cancer before symptoms appear.

Recommended screening methods include:

Stool-Based Tests

- **Fecal Occult Blood Test (FOBT)**
- **Fecal Immunochemical Test (FIT)**
- **Stool DNA Test**

Visual Examinations

- **Colonoscopy**
- **Flexible Sigmoidoscopy**
- **CT Colonography**

Individuals at average risk should begin screening according to national healthcare guidelines.

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07. Colonoscopy

Colonoscopy is considered the gold standard for colorectal cancer screening.

It allows physicians to:

- **Detect polyps**
- **Remove polyps**
- **Collect tissue samples**
- **Diagnose early cancer**

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08. Diagnosis

Diagnosis involves several investigations.

Medical History

The doctor evaluates:

- **Symptoms**
- **Family history**
- **Lifestyle factors**
- **Previous bowel disease**

Physical Examination

Includes:

- **Abdominal examination**
- **Digital rectal examination**

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09. Imaging Tests

Imaging helps determine the size and spread of cancer.

Common imaging includes:

- **CT Scan**
- **MRI**
- **PET Scan**
- **Chest X-ray**

These tests assist in staging.

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10. Biopsy

A biopsy confirms the diagnosis.

During colonoscopy, tissue samples are collected and examined under a microscope.

Biopsy determines:

- **Cancer type**
- **Tumor grade**
- **Treatment planning**

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11. Cancer Staging

Colorectal cancer is divided into stages.

Stage I

Cancer remains within the colon wall.

Stage II

Cancer extends beyond the colon wall.

Stage III

Cancer has spread to nearby lymph nodes.

Stage IV

Cancer has spread to distant organs such as:

- **Liver**
- **Lungs**
- **Bones**

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12. Treatment Options

Treatment depends on:

- **Stage**
- **Tumor location**
- **Patient age**
- **Overall health**

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13. Surgery

Surgery is the primary treatment for localized colorectal cancer.

Procedures include:

Polypectomy

Removal of precancerous polyps.

Partial Colectomy

Removal of the cancerous section of the colon.

Total Colectomy

Removal of the entire colon when necessary.

Nearby lymph nodes may also be removed.

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14. Chemotherapy

Chemotherapy destroys cancer cells throughout the body.

It may be used:

- **After surgery**
- **Before surgery**
- **For advanced disease**

Common side effects include:

- **Fatigue**
- **Hair loss**
- **Nausea**
- **Diarrhea**
- **Infection risk**

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15. Radiation Therapy

Radiation therapy is mainly used for rectal cancer.

Benefits include:

- **Shrinking tumors before surgery**
- **Reducing recurrence**
- **Relieving symptoms in advanced disease**

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16. Targeted Therapy

Targeted medicines block specific molecules responsible for cancer growth.

Examples include therapies targeting:

- **VEGF pathway**
- **EGFR pathway**

These treatments are generally combined with chemotherapy.

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17. Immunotherapy

Immunotherapy is useful for selected patients whose tumors have specific genetic characteristics.

Benefits include:

- **Improved survival**
- **Better disease control**
- **Long-lasting responses**

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18. Possible Complications

Untreated colorectal cancer may lead to:

- **Intestinal obstruction**
- **Internal bleeding**
- **Severe anemia**
- **Liver metastasis**
- **Lung metastasis**
- **Malnutrition**

Prompt treatment reduces these risks.

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19. Prevention

Healthy lifestyle choices reduce colorectal cancer risk.

Recommendations include:

- **Eat high-fiber foods.**
- **Increase fruits and vegetables.**
- **Reduce processed meat intake.**
- **Limit red meat consumption.**
- **Exercise regularly.**
- **Maintain healthy body weight.**
- **Avoid smoking.**
- **Limit alcohol consumption.**
- **Participate in regular screening.**

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20. Nutrition During Treatment

Patients are encouraged to consume:

- **Soft foods**
- **Whole grains**
- **Fruits**
- **Vegetables**
- **Lean protein**
- **Plenty of fluids**

Foods causing bowel irritation should be limited if recommended by healthcare providers.

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21. Follow-Up Care

Regular follow-up helps monitor recovery and detect recurrence.

Patients should attend scheduled:

- **Colonoscopy**
- **Blood tests**
- **CT scans**
- **Physical examinations**

Any new bowel symptoms should be reported immediately.

22. Key Points

- **Colorectal cancer often develops from precancerous polyps.**
- **Colonoscopy is the most effective screening method.**
- **Blood in stool should never be ignored.**
- **Surgery remains the primary treatment.**
- **Chemotherapy, radiation therapy, targeted therapy, and immunotherapy are important treatment options.**
- **Healthy diet and regular screening greatly reduce risk.**
- **Early diagnosis significantly improves survival rates.**